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ABSTRACT

Online Quiz Application is a web-based assessment system designed to conduct examinations in a simple, secure, and efficient manner using modern web technologies. The rapid growth of digital education and e-learning platforms has created a strong demand for online examination systems that reduce manual effort, save time, and ensure accuracy. This project aims to develop an Online Quiz Application using HTML, CSS, and JavaScript to provide a user-friendly and automated quiz environment for students and educators. The application allows students to enter their personal details such as name, roll number, register number, and select the subject before attending the quiz. After reading the examination rules, students can start the quiz, where each question is displayed with multiple-choice options and a fixed time limit. A timer mechanism is implemented to control the duration of each question, ensuring time-bound assessment and improving fairness in evaluation. The system automatically moves to the next question when the timer expires. To maintain examination integrity, the application includes a malpractice detection feature using window focus monitoring. If a student switches tabs, minimizes the browser, or changes focus repeatedly, the system generates warnings and can automatically submit or block the quiz based on predefined limits. This feature helps reduce cheating and ensures a transparent assessment process. The Online Quiz Application provides a time-bound and automated examination process with instant result generation, reducing manual work and evaluation delay. The system ensures exam integrity by monitoring browser focus changes and controlling malpractice during the quiz session. The quiz results are generated automatically after submission, displaying the student's details, selected subject, score, and status. This eliminates the need for manual evaluation and provides instant feedback to students. The system is lightweight, does not require additional software installation, and can be accessed through any standard web browser. Overall, the Online Quiz Application provides an efficient, reliable, and cost-effective solution for conducting online assessments. It is suitable for colleges, training institutes, and educational organizations for internal tests, practice exams, and evaluations. The project demonstrates how basic web technologies can be effectively used to build a functional and secure online examination system.

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CHAPTER 1

INTRODUCTION

1.1 OVERVIEW OF THE PROJECT

The Online Quiz Application is a web-based project developed to conduct quizzes and assessments in a simple, efficient, and automated manner. The project is designed using basic web technologies such as HTML for structure, CSS for user interface design, and JavaScript for implementing application logic. It aims to replace traditional manual examination methods with a digital system that reduces paperwork, saves time, and improves accuracy in evaluation. The application can be accessed through a web browser, making it flexible and easy to use for both students and educators. In this system, students are required to enter their basic details such as name, roll number, register number, and select the subject before starting the quiz. After reading the examination rules, students can attend the quiz where multiple-choice questions are displayed one by one. Each question is associated with a fixed time limit, ensuring that the assessment is time-bound and fair for all participants. The system automatically manages question navigation and timing without manual intervention. The project also focuses on maintaining examination integrity by incorporating basic monitoring mechanisms. Browser window focus detection is used to identify tab changes or focus loss during the quiz. If such actions occur repeatedly, the system generates warnings and can automatically submit the quiz, thereby reducing the chances of malpractice. After the quiz is completed, the application evaluates the answers automatically and displays the result instantly, including the score and student details. Overall, the Online Quiz Application provides a reliable and practical solution for conducting online assessments in educational institutions.

1.2 NEED FOR THE PROJECT

In today's educational environment, the traditional method of conducting exams and assessments manually is often time-consuming, prone to errors, and lacks efficiency in evaluating large groups of students. There is a growing need for a digital solution that simplifies the assessment process while ensuring accuracy and transparency. An Online Quiz Application addresses this need by providing a platform where students can participate in quizzes from anywhere, while educators can monitor performance in real-time. It reduces paperwork, enables instant result generation, and prevents malpractices through features like automatic submission, timing restrictions, and tab-switching alerts. Additionally, it allows for customizable quizzes tailored to different subjects and difficulty levels, supports remote learning for students in different locations, and enhances student engagement through interactive interfaces and instant feedback. To ensure reliability, the system can store and backup all results securely, track user activity to prevent data loss or tampering, and maintain consistent performance under high user loads, making it a

dependable tool for both students and educators. Moreover, such a system facilitates better data management, allowing teachers to analyse student performance efficiently and make informed decisions for improving learning outcomes. Hence, the implementation of an online quiz application is essential to modernize assessments and make the evaluation process faster, secure, and more reliable.

1.3 OBJECTIVE OF THE PROJECT

The primary objective of the Online Quiz Application is to digitize and streamline the process of conducting quizzes and assessments. It aims to provide a reliable, efficient, and user-friendly platform for both students and educators. The system is designed to reduce the dependency on manual paper-based exams, minimizing the chances of errors and saving time in evaluation and result generation.

1)Another important objective is to ensure fairness and integrity in assessments. Features such as question timers, automatic submission, and tab-switching alerts help prevent malpractices, ensuring that every student is evaluated under consistent conditions. The project also focuses on instant feedback and performance tracking, enabling students to identify their strengths and weaknesses while allowing educators to analyse overall performance effectively.

2)Additionally, the project seeks to enhance accessibility and engagement by allowing students to attempt quizzes remotely from any device with internet access. It supports customizable quizzes for different subjects and difficulty levels, promoting a more personalized learning experience. Finally, the system is built with secure data management and reliability in mind, ensuring that all results and user information are stored safely and backed up, making it a dependable tool for educational institutions.

1.4 SCOPE OF THE PROJECT

The scope of this project focuses on designing and developing a web-based Online Quiz Application that supports digital assessment in an efficient and systematic manner. The project aims to automate the entire quiz process, reducing manual effort and improving accuracy in evaluation and result generation. This project covers the complete assessment workflow, including student registration, quiz participation, monitoring, evaluation, and result management. By implementing structured modules, the system ensures fairness, transparency, and consistency in online examinations.

1. Student Registration and Quiz Participation

The Student Registration and Quiz Participation module plays a crucial role in the Online Quiz Application, as it serves as the entry point for users accessing the system. This subtopic focuses on collecting and managing essential student information in a structured and secure manner. Students are required to register by

entering details such as name, roll number, registration number, and subject selection. This process ensures proper identification of each participant and avoids duplication or unauthorized access. Once registration is completed, students are allowed to proceed to the quiz section. Clear instructions and rules are displayed before the quiz begins to ensure that students understand the assessment guidelines. The system provides a smooth transition from registration to quiz participation, minimizing confusion and improving user experience. Students can attempt quizzes from any location using a web browser, which increases accessibility and supports remote learning. The quiz participation process is designed to be simple and user-friendly. Questions are displayed clearly along with multiple-choice options, making it easy for students to select answers. Navigation controls allow students to move through questions systematically. The system ensures that responses are recorded accurately and securely until final submission. This subtopic also includes validation mechanisms to ensure that mandatory fields are filled correctly before starting the quiz. By implementing proper registration and participation flow, the system guarantees fairness, accountability, and smooth operation throughout the assessment process. Overall, this module ensures that students can participate in online assessments confidently and without technical difficulties.

2. Time Management and Malpractice Monitoring

Time management is a critical aspect of any assessment system, and this project places strong emphasis on implementing effective time-based controls. Each quiz or question is assigned a predefined time limit to ensure uniform assessment conditions for all students. A countdown timer is displayed on the quiz screen, allowing students to manage their time efficiently while answering questions. When the allotted time expires, the system automatically submits the quiz, eliminating the need for manual intervention. This subtopic also focuses on monitoring user behaviour to reduce malpractice during online examinations. Since online assessments lack physical supervision, the system includes basic monitoring features such as detecting tab switching or moving away from the quiz window. When such activities are detected, warning messages are displayed to alert the student. These measures help discourage unfair practices and maintain the credibility of the examination process. Additionally, the system can restrict certain actions such as page refresh or right-click operations during the quiz. These controls ensure that students remain focused on the assessment and do not attempt to manipulate the system. All monitoring activities are logged, allowing teachers or administrators to review suspicious behaviour if required. By combining time constraints with monitoring mechanisms, this module ensures fairness, discipline, and transparency in online assessments. It helps create a controlled examination environment similar to traditional classroom tests while leveraging the advantages of digital platforms.

3. Automated Evaluation and Result Management

Automated evaluation and result management form one of the most important components of the Online Quiz Application. This subtopic focuses on eliminating manual correction by automatically evaluating student responses once the quiz is submitted. The system compares submitted answers with predefined correct answers and calculates scores instantly. This significantly reduces evaluation time and avoids human errors associated with manual grading. After evaluation, results are generated immediately and displayed to students, providing instant feedback on their performance. This helps students understand their strengths and weaknesses without delay. For teachers, the system stores all results in a structured format, making it easy to access, review, and analyses student performance over time. Result management also includes maintaining records of multiple quiz attempts, subjects, and scores. These records can be used to generate reports for academic evaluation and progress tracking. The system ensures that result data is stored securely and protected from unauthorized access or modification. Backup mechanisms further enhance reliability by preventing data loss. This module greatly reduces administrative workload and improves efficiency in the assessment process. By automating evaluation and result handling, the system provides a fast, accurate, and reliable method of managing academic assessments, making it highly suitable for educational institutions.

CHAPTER 2

LITERATURE SURVEY

Traditional examination systems

Traditional examination systems are widely used in educational institutions and mainly on paper-based tests. In this method, question papers are printed, distributed to students, and answers are written manually. Evaluation is performed by teachers, which consumes a significant amount of time and effort. Literature shows that manual examination systems are prone to human errors, delays in result declaration, and difficulties in handling large volumes of answer scripts. Several studies highlight that traditional systems lack flexibility and scalability. Conducting exams for a large number of students requires extensive manpower and physical infrastructure. Additionally, storing and maintaining records manually increases the risk of data loss and mismanagement. These limitations have motivated researchers to explore digital alternatives for assessment.

Computer-Based Testing Systems

Computer-Based Testing (CBT) systems were introduced to overcome the drawbacks of manual examinations. In CBT, students attempt exams on computers, and answers are evaluated electronically. Research indicates that CBT systems reduce evaluation time and improve accuracy in scoring. These systems also support objective-type questions such as multiple-choice questions, which are suitable for automated assessment. However, literature also points out certain limitations of early CBT systems. Many systems required specialized software and hardware, limiting accessibility. Some lacked proper security mechanisms, making them vulnerable to malpractice. These findings emphasize the need for web-based solutions that are easily accessible and secure.

Web-Based Online Quiz Applications

Web-based online quiz applications have gained popularity due to their accessibility and ease of use. Studies show that such systems allow students to attempt quizzes from any location using a web browser, supporting remote learning environments. These applications are commonly developed using web technologies like HTML, CSS, and JavaScript, which makes them platform-independent. Literature highlights that web-based quiz systems provide features such as instant result generation, automated evaluation, and centralized data storage. They also reduce administrative workload and enhance user experience. However, some existing systems lack advanced monitoring features and robust data security, indicating scope for improvement.

Security and Malpractice Prevention in Online Assessments

Security is a major concern in online examination systems. Various research papers discuss challenges such as impersonation, tab switching, unauthorized access, and data manipulation. To address these issues, different techniques like user authentication, activity tracking, time restrictions, and browser control mechanisms have been proposed. Literature suggests that implementing monitoring features such as tab change detection and automatic submission can significantly reduce malpractice. However, no single solution can completely eliminate cheating, and hence a combination of preventive measures is recommended. These studies influence the design of secure and reliable online quiz applications.

Automated Evaluation and Performance Analysis

Automated evaluation is one of the most significant advancements discussed in literature related to online assessments. Research indicates that automated grading systems provide fast and accurate results, eliminating human bias and errors. These systems are particularly effective for objective-type questions and can handle large numbers of submissions efficiently. Performance analysis features such as score tracking, result storage, and progress monitoring are also emphasized in several studies. Literature shows that analysing student performance over time helps educators identify learning gaps and improve teaching strategies. These insights support the inclusion of result management and analytics modules in online quiz application

CHAPTER 3

SYSTEM DESIGN

3.1 Proposed System Architecture Design

The proposed system architecture design of the Online Quiz Application describes the overall structure and working flow of the system components. The system is designed as a web-based application that can be accessed through a standard web browser. It is developed using HTML, CSS, and JavaScript, which together provide structure, styling, and functional behaviour to the application. In the proposed architecture, the user interface acts as the primary interaction layer where students enter their details, read instructions, attempt quiz questions, and view results. JavaScript handles the application logic such as question sequencing, time management, response validation, and automatic submission of the quiz. All operations are executed at the client side, ensuring fast response and smooth interaction. The system architecture also includes basic monitoring mechanisms to support secure assessment. Events such as tab switching, window focus change, and page refresh attempts are detected during the quiz session. When such activities occur, the system generates warning messages and logs the actions to maintain examination discipline. The proposed system architecture is simple, lightweight, and efficient for college-level assessments. It supports easy maintenance and future scalability. Additional components such as database integration, server-side processing, and teacher administration modules can be incorporated into this architecture in future enhancements.

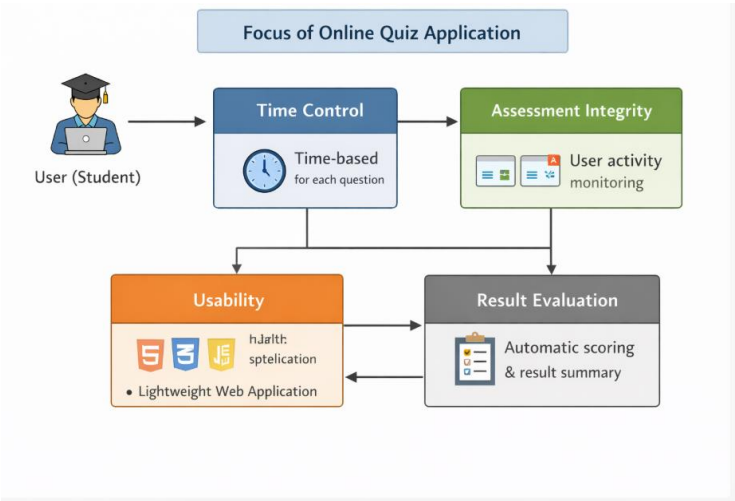


FIG.3.1. PROPOSED ARCHITECTURE

3.1 MODULE DESIGN

1. User Management

The User Management module is the backbone of the online quiz system as it manages all participants and ensures system security. It allows students to register by providing key details such as their name, roll number, registration number, email, and password. These details help uniquely identify each student and prevent duplicate entries. Staff members, such as teachers or administrators, are provided with secure login credentials that allow them to manage quizzes, view results, and monitor quiz activity. This module also provides profile management features, enabling users to update personal information and change their passwords whenever necessary. Authentication ensures that only authorized users can access the system, maintaining the integrity of the data. For example, if a student tries to log in without registering, the system will block access, ensuring security. Overall, this module ensures smooth user registration, secure access, and proper identity management.

2. Subject & Quiz Management

The Subject & Quiz Management module is responsible for organizing the quiz content and making it accessible to students. Students can select their desired subjects, such as HTML, CSS, or JavaScript, based on their curriculum. Staff members can create quizzes for each subject by adding questions, defining multiple-choice options, marking correct answers, and setting time limits for each question. Additionally, staff can edit or delete existing quizzes to ensure that content remains accurate and updated. The system supports different types of questions such as multiple-choice and true/false, which allows flexibility in designing quizzes for different learning objectives. This module ensures that both students and staff can interact with quizzes in an organized and systematic manner.

3. Quiz Rules & Instructions

Before a student begins a quiz, the Quiz Rules & Instructions module ensures that they understand the guidelines and requirements. This module displays instructions such as quiz duration, how to navigate between questions, and the number of allowed attempts. Typically, only one attempt per student is allowed to maintain fairness. The module also includes a warning system, which alerts students if they switch tabs or attempt to refresh the page. Such measures prevent cheating and ensure proper conduct. For example, if a student moves to another browser tab, a pop-up warning can appear reminding them that the quiz may end automatically if the rules are violated. This module is essential for maintaining discipline and fairness during the quiz.

4.Quiz Attempt & Timer

The Quiz Attempt & Timer module manages the actual process of taking the quiz. Each question is accompanied by a countdown timer, and the system automatically moves to the next question once the timer ends. This ensures that students answer questions within the allocated time. To maintain fairness, this module also disables right-click functionality and page refresh, preventing students from attempting to cheat or manipulate the system. If the student violates any rules or the quiz time expires, the system automatically submits the quiz. This module ensures smooth, regulated, and fair quiz attempts, reducing manual intervention by staff.

5. Result & Scoring

The Result & Scoring module is responsible for evaluating quiz submissions. Once the quiz is submitted, the system automatically calculates the score based on the number of correct answers. Results are displayed instantly to students, providing immediate feedback on their performance. Additionally, results are stored in the database for future reference. Staff can view these results to track student progress over time. An optional feature of this module is a leaderboard, which displays top-performing students, motivating learners to improve their performance. This module ensures transparency, efficiency, and instant feedback in the quiz system.

6. Admin/Staff Monitoring

The Admin/Staff Monitoring module provides tools for staff to oversee quizzes and student activity. Staff can monitor quizzes in real-time, detect potential malpractice, and ensure that students are following rules. The system can track activities like tab switching or multiple attempts, alerting staff to possible cheating. Administrators can also generate reports, view and download student scores, and manage quiz questions by adding, editing, or deleting them. This module ensures smooth administration, accountability, and proper supervision, allowing staff to maintain a secure and effective online quiz environment

CHAPTER 4

REQUIREMENTS SPECIFICATION

4.1 HARDWARE REQUIREMENTS

The hardware requirements are fundamental for ensuring that the Focus Online Quiz Application functions smoothly and reliably. Since this system is web-based, it depends on both client-side devices for students and staff, as well as server-side infrastructure for hosting, managing quizzes, and storing results. Adequate hardware ensures fast performance, accurate timing, real-time interactions, and uninterrupted quiz experiences. On the client side, students and staff need devices with a minimum Intel i3 processor and 4 GB RAM to handle browser operations efficiently. Sufficient storage of at least 100 MB free space is required for temporary files, browser cache, and offline data storage if needed. A monitor or laptop with a resolution of 1024x768 pixels ensures that quiz questions, options, and timer are displayed clearly. Functional input devices such as a keyboard, mouse, or touchpad are essential for accurate and timely responses. A stable internet connection of at least 1 Mbps is necessary to support real-time quiz submission, timer synchronization, and alert notifications. Smooth client-side performance is essential to prevent lag, page freezes, or errors that could disrupt the quiz experience. On the server side, hosting the application requires robust hardware to support multiple users simultaneously. A minimum Intel i5 processor with multi-core capability and 8 GB RAM is recommended to manage concurrent quiz attempts, database operations, and background tasks efficiently. Storage requirements include at least 500 GB HDD or 256 GB SSD to store quizzes, student data, and results; SSDs are preferred for faster read/write operations and better performance under load. High-speed internet connectivity of 10 Mbps or higher with low latency is essential for seamless communication between clients and the server, especially during real-time quizzes. Regular backup facilities, such as cloud storage or external drives, are crucial to prevent data loss and maintain system reliability. The server operating system should be compatible with web servers like Apache, Nginx, or IIS to ensure smooth application deployment. Peripheral devices also play an important role in supporting the quiz environment. A UPS (Uninterruptible Power Supply) ensures continuous power supply and prevents sudden shutdowns during exams. Printers or scanners may be required for printing reports or scanned answer sheets. Networking devices, such as routers, switches, and modems, ensure stable internet connectivity across the institution network. These components collectively enhance the performance, reliability, and monitoring of the system. In conclusion, having adequate hardware ensures that the Focus Online Quiz Application operates efficiently, maintains real-time interactions, provides accurate timing for each question, and stores results securely. Meeting both client-side and server-side requirements minimizes technical issues, guarantees

smooth functioning during high-stakes quizzes, and enhances the overall user experience for students and staff.

4.2 SOFTWARE REQUIREMENTS

The software requirements define the programs, platforms, and tools necessary for the development, deployment, and smooth functioning of the Focus Online Quiz Application. These requirements ensure compatibility, security, and an interactive experience for both students and staff. The requirements are categorized into Client-Side Software, Server-Side Software, Database Management, and Supporting Tools & Libraries for clarity.

1. Client-Side Software

The client-side software enables students and staff to access the quiz application through web browsers. Users require modern browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari. These browsers must support HTML5, CSS3, and JavaScript to ensure that interactive quiz features, such as multiple-choice questions, countdown timers, and real-time alerts, function properly. Updated browsers are essential to prevent compatibility issues.
Additionally, the client devices should have a stable operating system such as Windows, Linux, or macOS to run the browser efficiently. Internet connectivity is crucial, as the quizzes require real-time synchronization between client devices and the server.

2. Server-Side Software

The server-side software is responsible for hosting the application, handling requests from multiple users, and processing quiz-related operations. The server must have a compatible operating system such as Windows Server, Linux (Ubuntu, CentOS), or macOS.
A web server software such as Apache, Nginx, or IIS is required to serve web pages, manage HTTP requests, and handle client-server communication. For backend logic, programming or scripting languages like JavaScript (Node.js), PHP, or Python are used to process quiz attempts, manage user authentication, and communicate with the database securely.

3. Database Management System (DBMS)

A reliable Relational Database Management System (RDBMS) such as MySQL, PostgreSQL, or SQLite is essential to store student details, quiz questions, options, responses, and results. Proper database management ensures that all data is stored securely, retrieved accurately, and can be used for generating reports or leaderboards. Regular backup and maintenance of the database are also part of the software requirements to prevent data loss.

4. Supporting Tools & Libraries

Several supporting tools enhance the functionality and usability of the quiz application. Integrated Development Environments (IDEs) like Visual Studio Code, Eclipse, or WebStorm are required for coding and debugging the application efficiently. Version control systems, such as Git or GitHub, help in tracking code changes, collaboration, and maintaining source code integrity.
Additional libraries and frameworks such as Bootstrap for responsive design, jQuery for dynamic content, and Chart.js for visualizing quiz results improve the overall user experience.

5. Connectivity & Updates

Stable internet connectivity is critical, as the application relies on real-time communication between client and server. Regular updates of web browsers, operating systems, server software, and database systems are necessary to maintain security, efficiency, and compatibility with modern web standards.

4.3 SOFTWARE SPECIFICATION

The software specification defines all software components, frameworks, tools, and standards required to design, develop, deploy, and maintain the Focus Online Quiz Application efficiently. It ensures the system is secure, interactive, and user-friendly, providing seamless functionality for both students and staff. Clear software specifications also help developers, administrators, and IT staff understand the required environment, dependencies, and technologies to implement the project effectively. The specifications cover client-side software, server-side software, database management, development tools, libraries and frameworks, standards, security, and connectivity.

1. Client-Side Software Specification

The client-side software enables students and staff to access the application through web browsers on desktops, laptops, or tablets. Modern browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari are required, as they support HTML5, CSS3, and JavaScript features necessary for rendering quiz questions, interactive options, countdown timers, real-time alerts, and responsive interfaces. Outdated browsers may fail to display content properly or break interactive features. Client devices must have stable operating systems such as Windows, Linux, or macOS. They should also have sufficient memory (minimum 4 GB RAM), processor speed (minimum Intel i3), and storage (minimum 100 MB free) to handle multiple browser tabs, scripts, and temporary files. A reliable internet connection is essential for real-time quiz submissions, timer synchronization, and displaying alerts. The system also requires secure communication protocols (HTTPS) to ensure that login credentials and

quiz data remain encrypted and safe during transmission. This client-side specification ensures that users can attempt quizzes smoothly without facing display issues, browser crashes, or lag during timed assessments. It guarantees a consistent and reliable user experience across different devices and operating systems.

2. Server-Side Software Specification

The server-side software is the backbone of the online quiz system, hosting the application, managing multiple user requests, processing quiz logic, and storing data securely. The server should run on compatible operating systems such as Windows Server, Linux (Ubuntu/CentOS), or macOS. It must support web server software like Apache, Nginx, or IIS to efficiently handle HTTP requests, serve web pages, and maintain connections with multiple clients simultaneously. The backend logic of the application can be implemented using programming languages such as Node.js, PHP, or Python, which process user authentication, quiz submissions, auto-scoring, timer management, and database communication. The server-side specification also includes security measures like firewalls, SSL encryption, secure password hashing, and protection against unauthorized access. Efficient handling of multiple concurrent quiz attempts is critical to ensure that the system performs reliably during peak usage. Additionally, the server must be capable of logging events such as user login/logout, quiz attempts, and alert notifications for administrators to monitor and audit activities. Automatic updates and maintenance scripts are recommended to keep the server software secure and optimized.

3. Database Specification

A well-designed database is crucial to store and manage data such as student profiles, quiz questions, multiple-choice options, responses, and results. The system requires a Relational Database Management System (RDBMS) like MySQL, PostgreSQL, or SQLite, which ensures data integrity, consistency, and security. The database should support efficient querying, fast retrieval of questions, real-time storage of quiz responses, and quick calculation of scores. Sensitive data, such as passwords, must be stored using encryption or hashing algorithms to prevent unauthorized access. The system must also support backup and recovery mechanisms, both manual and automated, to prevent data loss in case of server failure or other technical issues. Proper database normalization and indexing are necessary to maintain performance when handling large numbers of concurrent users. The database must be scalable to accommodate future additions, such as new subjects, quizzes, or users, without affecting performance.

4. Development Tools and IDEs

Efficient software development requires the use of Integrated Development Environments (IDEs) and supporting tools. Tools like Visual Studio Code, Eclipse, or WebStorm allow developers to write, debug, and test code effectively. Version control systems such as Git or GitHub are essential for managing source code, tracking changes, and enabling collaboration among team members. Package managers like nm. or Composer help manage dependencies, external libraries, and frameworks, ensuring that all required modules are installed and updated correctly. Testing tools, including browser developer tools and automated testing frameworks, are important to identify bugs, validate features, and maintain code quality.

5. Frameworks and Libraries

Frameworks and libraries are used to enhance functionality, interactivity, and responsiveness. Bootstrap provides a responsive design that works across devices and screen sizes. jQuery is used for dynamic content, animations, and handling user interactions such as button clicks or timer updates. Chart.js can be implemented to display graphical results and performance metrics. Validation libraries ensure that user inputs, such as registration data and quiz responses, are accurate and complete, preventing errors during submission. These frameworks and libraries reduce development time while enhancing the user experience and interface quality.

6. Standards and Security Specifications

The software must adhere to web standards, including HTML5, CSS3, and JavaScript ES6, to maintain compatibility across modern browsers. Security standards include SSL, HTTPS encryption, password hashing, and secure authentication protocols to protect sensitive data. Performance standards ensure optimized code, efficient server-side processing, and low latency during multiple concurrent quiz attempts. Regular updates of web browsers, operating systems, server software, and databases ensure stability, security, and compatibility with modern web technologies. Backup procedures, logging, and auditing further enhance reliability and accountability.

7. Connectivity and Real-Time Requirements

Stable internet connectivity is essential for real-time quiz operation, timer synchronization, and alert notifications. Both client and server software must support instant data transfer, allowing students to attempt quizzes and submit responses without delays. The server must handle multiple concurrent connections, and the database must process and store data in real-time to ensure accuracy of scoring and res

5.CHAPTER

IMPLEMENTATION

5.1 SAMPLE CODE

```
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Advanced Online Quiz with Behaviour Tracking</title>
<style>
body{font-family:Arial; background:#eef2f9;}

/* CONTAINER STYLES */
.container{
  width:500px;
  margin:20px auto;
  padding:20px;
  border-radius:12px;
  box-shadow: 0 8px 20px rgba(0,0,0,0.2);
  transition: all 0.3s ease;
  color: #333;
}

/* BOX COLOURS */
#loginBox { background: linear-gradient(135deg, #e0c3fc, #8ec5fc); }
#rulesBox { background: linear-gradient(135deg, #f9d5ff, #c3cfff); }
#quizBox { background: linear-gradient(135deg, #ffffff, #d9e6ff); } /* quiz box soft
white/light blue */
```

```
#resultBox { background: linear-gradient(135deg, #d9f0ff, #ffffff); }
#staffBox { background: linear-gradient(135deg, #e0ffe0, #ffffff); }

.hidden{display:none;}

/* INPUTS & BUTTONS */
input,select,button{width:100%; padding:10px; margin-top:8px; border-radius:6px; border:1px solid #ccc;}
button{background:#2563eb;color:white;border:none;cursor:pointer; font-weight:bold; transition: 0.3s;}
button:hover{background:#1e4bb8;}
button:disabled{background:gray; cursor:not-allowed;}

/* WARNINGS & TIMER */
.warning{color:red;font-weight:bold;}
.timer{font-weight:bold;}

/* TABLE */
table{width:100%; border-collapse:collapse; margin-top:10px; background:#fff;}
th, td{border:1px solid #ccc; padding:6px; text-align:center;}
</style>
</head>
<body>
<!-- LOGIN -->
<div class="container" id="loginBox">
<h2>Quiz Login</h2>
<input type="text" id="name" placeholder="Enter Name">
<input type="text" id="roll" placeholder="Roll No">
<input type="text" id="reg" placeholder="Register No">
```

```
<select id="subject">
  <option value="">Select Subject</option>
  <option>Web Technology</option>
  <option>JavaScript</option>
  <option>Computer Basics</option>
</select>
<button onclick="goRules()">Continue</button>
</div>
<!-- RULES -->
<div class="container hidden" id="rulesBox">
<h3>Quiz Rules</h3>
<ul>
<li>No tab switching</li>
<li>Submit each question before next</li>
<li>Focus score decreases on malpractice</li>
<li>Timer per question (15 sec)</li>
<li>Random attention check questions may appear</li>
<li>Random or very fast answers reduce focus</li>
</ul>
<button onclick="startQuiz()">Accept & Start Quiz</button>
</div>
<!-- QUIZ -->
<div class="container hidden" id="quizBox">
<h4 id="qno"></h4>
<p id="question"></p>
<div id="options"></div>
<p>Focus Score: <span id="focus">100</span></p>
<p class="timer">Time Left: <span id="timer">15</span>s</p>
```

```
<p class="warning" id="warn"></p>

<button id="submitBtn" onclick="submitAnswer()">Submit</button>

<button id="nextBtn" onclick="nextQuestion()" disabled>Next</button>

</div>

<!-- RESULT -->

<div class="container hidden" id="resultBox">

<h2>Result</h2>

<p><b>Name:</b> <span id="rname"></span></p>

<p><b>Roll No:</b> <span id="rroll"></span></p>

<p><b>Register No:</b> <span id="rreg"></span></p>

<p><b>Subject:</b> <span id="rsub"></span></p>

<p><b>Score:</b> <span id="rscore"></span></p>a

<p><b>Focus Score:</b> <span id="rfocus"></span></p>

<p><b>Status:</b> <span id="rstatus"></span></p>

<button onclick="showStaff()">View Staff Summary</button>

</div>

<!-- STAFF VIEW -->

<div class="container hidden" id="staffBox">

<h2>Staff View - All Attempts</h2>

<table>

<tr><th>Name</th><th>Roll</th><th>Subject</th><th>Score</th><th>Focus</th><th>
Status</th></tr>

<tbody id="staffTable"></tbody>

</table>

<button onclick="location.reload()">Back / New Student</button>

</div>

<script>

// QUESTIONS
```

```
let questions=[
 {q:"HTML stands for?", o:["Hyper Text Markup Language","High Text","None"], a:0},
 {q:"CSS is used for?", o:["Logic","Styling","Database"], a:1},
 {q:"JS is a?", o:["Markup","Programming","Design"], a:1}
];
let attentionQuestions=[{q:"Attention Check: Select option C only", o:["A","B","C","D"],
a:2}];

let i=0, score=0, submitted=false, blocked=false, focusScore=100, timerInterval,
timeLeft=15;
let lastQuestionTime;

// LOGIN → RULES
function goRules(){
  const n=document.getElementById("name").value.trim();
  const r=document.getElementById("roll").value.trim();
  const reg=document.getElementById("reg").value.trim();
  const sub=document.getElementById("subject").value;
  if(!n || !r || !reg || !sub){ alert("Fill all details"); return;}
  loginBox.classList.add("hidden");
  rulesBox.classList.remove("hidden");
}

// RULES → QUIZ
function startQuiz(){
  rulesBox.classList.add("hidden");
  quizBox.classList.remove("hidden");
  loadQuestion();
}
```

```
// LOAD QUESTION
function loadQuestion(){
  submitted=false;
  submitBtn.disabled=false;
  nextBtn.disabled=true;
  warn.innerText="";
  lastQuestionTime = Date.now();

  if(Math.random() < 0.3){
    questions.splice(i,0,attentionQuestions[0]);
  }

  qno.innerText="Question "+(i+1);
  question.innerText=questions[i].q;
  options.innerHTML="";
  questions[i].o.forEach((op, idx)=>{
    options.innerHTML+=`<input type="radio" name="opt" value="${idx}"> ${op}<br>`;
  });
  startTimer();
}

// TIMER
function startTimer(){
  clearInterval(timerInterval);
  timeLeft=15;
  document.getElementById("timer").innerText=timeLeft;
  timerInterval=setInterval(()=>{
```

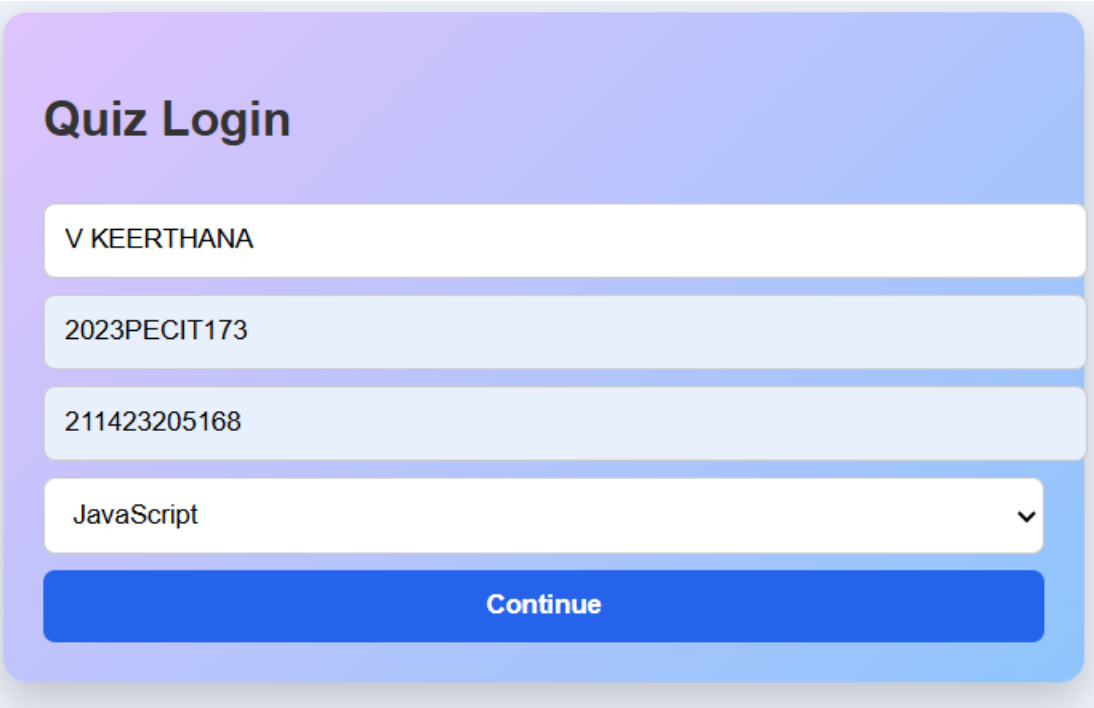
```
    timeLeft--;
    document.getElementById("timer").innerText=timeLeft;
    if(timeLeft<=0){
      clearInterval(timerInterval);
      submitAnswer();
      nextBtn.disabled=false;
    }
  },1000);
}

// SUBMIT ANSWER
function submitAnswer(){
  if(submitted) return;
  let sel=document.querySelector("input[name='opt']:checked");
  let answerTime = (Date.now() - lastQuestionTime)/1000;

  if(answerTime < 2) { focusScore -=5; warn.innerText = "Fast answer detected (-5 focus)"; }

  if(!sel){ focusScore -=5; warn.innerText = "No answer selected / Random answer (-5 focus)"; }
  else if(parseInt(sel.value)===questions[i].a){ score++; }
  else if(questions[i].q.includes("Attention Check")){ focusScore -=10; warn.innerText = "Attention check failed (-10 focus)"; }

  document.getElementById("focus").innerText = focusScore;
  submitted=true;
  submitBtn.disabled=true;
  nextBtn.disabled=false;
```

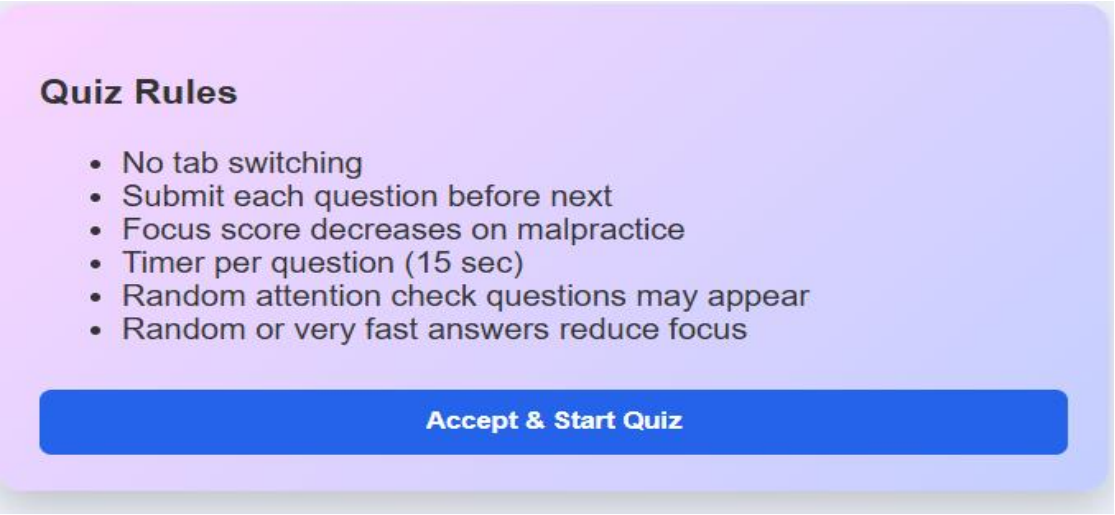
```
  clearInterval(timerInterval);
  if(focusScore<50) finishQuiz("Blocked - Malpractice");
}

// NEXT QUESTION
function nextQuestion(){
  if(!submitted) return;
  i++;
  if(i<questions.length){ loadQuestion(); }
  else{ finishQuiz("Completed"); }
}

// FINISH QUIZ
function finishQuiz(status){
  quizBox.classList.add("hidden");
  resultBox.classList.remove("hidden");
  rname.innerText=document.getElementById("name").value;
  rroll.innerText=document.getElementById("roll").value;
  rreg.innerText=document.getElementById("reg").value;
  rsub.innerText=document.getElementById("subject").value;
  rscore.innerText=score+" / "+questions.length;
  rfocus.innerText=focusScore;
  rstatus.innerText=status;
  blocked=true;
  clearInterval(timerInterval);

  let attempts=JSON.parse(localStorage.getItem("quizAttempts")||"[]");
  attempts.push({
```

```
    name:rname.innerText,
    roll:rroll.innerText,
    reg:rreg.innerText,
    subject:rsub.innerText,
    score:score,
    focus:focusScore,
    status:status
  });
  localStorage.setItem("quizAttempts", JSON.stringify(attempts));
}

// MALPRACTICE
document.addEventListener("visibilitychange",()=>{ if(document.hidden && !blocked){ focusScore -= 30; warn.innerText = "Warning! Tab switch detected (-30 focus)"; document.getElementById("focus").innerText = focusScore; if(focusScore<50) finishQuiz("Blocked - Malpractice"); } });
window.onblur=()=>{ if(!blocked){ focusScore -= 10; warn.innerText = "Warning! Window changed (-10 focus)"; document.getElementById("focus").innerText = focusScore; if(focusScore<50) finishQuiz("Blocked - Malpractice"); } };

// STAFF VIEW
function showStaff(){
  resultBox.classList.add("hidden");
  staffBox.classList.remove("hidden");
  let attempts=JSON.parse(localStorage.getItem("quizAttempts")||"[]");
  let tbody=document.getElementById("staffTable");
  tbody.innerHTML="";
  attempts.forEach(a=>{
    let tr=document.createElement("tr");
```

```
tr.innerHTML=`<td>${a.name}</td><td>${a.roll}</td><td>${a.subject}</td><td>${a.score}</td><td>${a.focus}</td><td>${a.status}</td>`;
   tbody.appendChild(tr);
 });
}
</script>
</body>
</html>
```

5.2 SAMPLE SCREEN SHOTS





Question 1

Attention Check: Select option C only

A

B

C

D

Focus Score: 90

Time Left: 10s

Attention check failed (-10 focus)

Submit

Next

Question 2

CSS is used for?

Logic

Styling

Database

Focus Score: 50

Time Left: 5s

Warning! Window changed (-10 focus)

Submit

Next

Question 3

CSS is used for?

- Logic
- Styling
- Database

Focus Score: 50

Time Left: 7s

Submit

Next

Result

Name: V KEERTHANA

Roll No: 2023PECIT173

Register No: 211423205168

Subject: Web Technology

Score: 2 / 3

Focus Score: 40

Status: Blocked - Malpractice

View Staff Summary

Staff View - All Attempts

Name	Roll	Subject	Score	Focus	Status
V KEERTHANA	2023PECIT173	Web Technology	1	50	Completed
V KEERTHANA	2023PECIT173	Web Technology	2	90	Completed
V KEERTHANA	2023PECIT173	JavaScript	1	100	Completed
V KEERTHANA	2023PECIT173	Computer Basics	2	40	Blocked - Malpractice
V KEERTHANA	2023PECIT173	Computer Basics	0	70	Completed
V KEERTHANA	2023PECIT173	Web Technology	0	100	Completed
V KEERTHANA	2023PECIT173	Web Technology	1	65	Completed
V KEERTHANA	2023PECIT173	JavaScript	1	90	Completed
V KEERTHANA	2023PECIT173	Computer Basics	0	45	Blocked - Malpractice
V KEERTHANA	2023PECIT173	Computer Basics	1	55	Completed
V KEERTHANA	2023PECIT173	Computer Basics	2	90	Completed

CHAPTER 6

TESTING AND MAINTANANCE

Testing is a fundamental phase in the software development life cycle, ensuring that software operates accurately, reliably, and securely. In the case of the Focused Online Quiz Application with Behaviour Tracking, testing goes beyond validating basic functionality. The system monitors student behavior in real time, including tab switching, attention check validation, rapid responses, and window focus loss, all of which can influence focus score and overall assessment results. Testing ensures that these features function correctly, providing a fair, accurate, and reliable platform for both students and faculty. By identifying errors and validating proper module interactions early, testing guarantees the integrity of the system while maintaining a smooth user experience.

1. Unit Testing

Unit testing focuses on testing individual modules of the application independently to verify that each unit performs its intended function correctly. In the Focused Online Quiz Application, each module such as login, rules acceptance, quiz display, timer, focus score calculation, and result generation is tested separately. The login module is tested to ensure that students cannot proceed without entering valid personal details and selecting a subject. The rules module is tested to confirm that students must accept the rules before starting the quiz. The quiz module is tested to verify correct question loading, option selection, and answer submission. Special attention is given to the timer unit to ensure that it starts correctly, counts down accurately, and triggers automatic submission when time expires. The focus score unit is tested under different scenarios such as fast answering, no answer selection, attention check failure, and tab switching. Unit testing helps isolate errors at an early stage, making debugging easier and ensuring each module is stable before integration.

2. Integration Testing

Integration testing ensures that all individual modules of the system work together smoothly as a complete application. In this project, integration testing verifies the interaction between the login module, quiz engine, timer system, focus score logic, and result display module. After a student logs in, their details must flow correctly to the quiz module and later to the result and staff view modules. Integration testing checks whether focus score changes during the quiz are immediately reflected in the user interface and stored correctly for final evaluation. The integration between the timer and auto-submit functionality is tested to ensure that unanswered questions are handled properly when time expires. The staff view module is also tested to confirm that all student attempts are retrieved correctly from local storage and displayed accurately.

This testing phase ensures data consistency, smooth navigation, and reliable communication between all components of the system.

3. Functional Testing

Functional testing validates whether the application behaves according to the specified functional requirements. In the Focused Online Quiz Application, functional testing verifies that all features provided to users operate correctly. This includes validating subject selection, quiz navigation, question loading, option selection, and submission workflow. Functional testing also ensures that warnings are displayed when students switch tabs, lose focus, or attempt to violate quiz rules. The correctness of focus score deduction rules is thoroughly tested to confirm that penalties are applied only under valid conditions. The result generation feature is tested to ensure that the final score, focus score, and quiz status are calculated and displayed accurately. Functional testing ensures that the system fulfills its intended purpose and delivers expected outputs for all valid and invalid inputs.

4. Performance Testing

Performance testing evaluates how efficiently the application performs under normal and heavy usage conditions. For the Focused Online Quiz Application, performance testing checks the responsiveness of the user interface, the speed of question loading, and the accuracy of the timer even during continuous usage. Multiple quiz attempts are simulated to ensure that local storage can handle repeated data storage without performance degradation. The application is tested across different browsers to ensure consistent performance. Performance testing also verifies that focus score updates and warning messages appear instantly without delay. This testing phase ensures that students experience a smooth, uninterrupted quiz environment, which is crucial for maintaining concentration and fairness during assessments.

5. Security and Behaviour Testing

Security and behaviour testing are vital in an online quiz system that aims to prevent malpractice. This type of testing focuses on validating the system's ability to detect and handle suspicious behavior such as tab switching, window changes, rapid guessing, and ignoring attention check questions. Behaviour testing ensures that focus score deductions are triggered correctly for each violation and that warning messages are clearly displayed to the user. Security testing also checks that quiz data cannot be manipulated easily and that students cannot bypass restrictions through browser actions. Automatic blocking of students whose focus score drops below the threshold is thoroughly tested to ensure fair enforcement of rules. This testing ensures academic integrity and builds trust in the system.

6. User Acceptance Testing

User Acceptance Testing ensures that the application meets the expectations of real users such as students and faculty members. In this phase, the application is tested in a real or simulated classroom environment. Students evaluate ease of use, clarity of instructions, visibility of timer and warnings, and overall quiz experience. Faculty members verify the accuracy of results, effectiveness of behaviour tracking, and usefulness of the staff summary view. Feedback collected during UAT is used to refine the interface, improve warning clarity, and optimize behaviour monitoring logic. This testing phase ensures that the application is practical, user-friendly, and suitable for academic deployment.

Test Cases Example

Test Case ID	Module	Scenario	Input	Expected Output	Status
TC1	Login	Empty Name	""	Alert: Fill all details	Pass
TC2	Quiz Timer	Time expires	15 sec	Auto-submit answer	Pass
TC3	Focus Score	Tab switch	Switch tab	Focus -30	Pass
TC4	Attention Check	Wrong option	Option != C	Focus -10	Pass

MAINTANANCE

Software maintenance refers to the activities required to keep a system operational, efficient, and relevant after deployment. In the Focused Online Quiz Application, maintenance is essential because academic requirements, technologies, and user expectations change over time. Maintenance ensures that the application continues to function correctly even when browsers are updated or new assessment rules are introduced. Since the system deals with sensitive student evaluation data, regular maintenance is required to preserve accuracy, security, and reliability. Effective maintenance helps extend the lifespan of the application while minimizing errors and ensuring consistent performance.

1. Corrective Maintenance

Corrective maintenance involves fixing errors or bugs that are identified after the system is deployed. In this application, issues such as incorrect focus score calculation, timer glitches, or result display errors may arise during real usage. Corrective maintenance focuses on identifying the root cause of these problems, modifying the code, and thoroughly retesting the affected modules. Quick corrective actions ensure that students are not unfairly evaluated and that faculty can trust the

system's results. This type of maintenance plays a key role in maintaining system credibility and user confidence.

2. Adaptive Maintenance

Adaptive maintenance is performed to keep the application compatible with changes in the operating environment. This includes adapting the system to new browser versions, updated academic policies, or different institutional requirements. For example, timer durations may need adjustment, new subjects may be added, or focus score thresholds may be modified. Adaptive maintenance ensures that the system remains usable and effective even as external conditions evolve. This flexibility allows the application to be deployed across different institutions without major redevelopment.

3. Perfective Maintenance

Perfective maintenance focuses on improving the system's performance, usability, and features based on user feedback and evolving needs. Enhancements such as better user interface design, more detailed staff analytics, improved warning messages, and optimized behaviour tracking algorithms fall under perfective maintenance. In the Focused Online Quiz Application, perfective maintenance ensures that the system becomes more efficient, user-friendly, and effective over time. These improvements increase user satisfaction and enhance the overall quality of the application.

4. Preventive Maintenance

Preventive maintenance aims to reduce the likelihood of future failures by improving system structure and code quality. This includes refactoring code, removing redundant logic, updating documentation, and optimizing performance-critical sections. Preventive maintenance also involves reviewing security measures and behaviour tracking logic to ensure long-term reliability. By proactively addressing potential issues, preventive maintenance helps reduce downtime, improve stability, and simplify future enhancements.

5. Maintenance Strategy and Future Scope

A well-defined maintenance strategy ensures systematic updates, regular testing, and proper version control. Scheduled maintenance checks, data backup strategies, and performance monitoring are essential to keep the system stable. Future enhancements such as AI-based behaviour analysis, advanced analytics dashboards, and mobile compatibility can be incorporated through effective maintenance planning. This approach ensures that the Focused Online Quiz Application remains scalable, secure, and relevant for future academic needs

CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENT

7.1 CONCLUSION

The Focused Online Quiz Application with Behaviour Tracking has been successfully designed and implemented to address the growing need for secure, reliable, and fair online assessment systems in modern educational environments. With the rapid adoption of digital learning platforms, traditional examination methods face significant challenges such as malpractice, lack of monitoring, and inconsistent evaluation standards. This project effectively overcomes these limitations by integrating behaviour tracking mechanisms, focus score evaluation, attention check questions, and real-time monitoring features that ensure academic integrity throughout the assessment process.The system provides a structured and user-friendly workflow starting from student login, rule acknowledgement, timed quiz participation, and automated result generation. By enforcing strict quiz rules and monitoring user behaviour such as tab switching, window focus changes, and abnormal answering patterns, the application ensures that students remain attentive and compliant during the assessment. The introduction of focus score calculation adds an additional layer of evaluation beyond traditional scoring, allowing instructors to assess not only academic performance but also student attentiveness and honesty.From a technical perspective, the application demonstrates effective integration of HTML, CSS, and JavaScript to create an interactive, responsive, and functional system. Local storage is utilized efficiently to store student attempts and generate staff summaries without requiring complex backend infrastructure. Comprehensive testing and structured maintenance strategies ensure system stability, reliability, and long-term usability. Overall, this project achieves its objectives by delivering a robust online quiz platform that enhances fairness, transparency, and trust in digital assessments. It serves as a practical and scalable solution suitable for academic institutions seeking secure online evaluation methods.

7.2 FUTURE ENHANCEMENT

Although the Focused Online Quiz Application with Behaviour Tracking successfully fulfills its current objectives, there is significant scope for future enhancements to further improve functionality, security, and user experience. One of the most promising enhancements is the integration of artificial intelligence–based behaviour analysis. AI algorithms can be used to analyze student interaction patterns more accurately, detecting subtle forms of malpractice such as repeated guessing, abnormal response timing, or unusual navigation behaviour.Another important enhancement is the incorporation of webcam-based online proctoring. By enabling face detection, eye movement tracking, and environment monitoring, the system can provide higher levels of exam security, making it suitable for high-stakes examinations. The application can

also be extended with a backend database and authentication system to support large-scale deployments across multiple institutions, enabling centralized data storage, secure login credentials, and role-based access control for administrators and faculty.Future versions of the system may include adaptive question difficulty based on student performance, providing personalized assessments that adjust in real time. Advanced analytics dashboards for faculty can be introduced to visualize student behaviour trends, focus scores, and performance insights. Additionally, mobile responsiveness and native mobile application support can be implemented to allow students to attend quizzes on smartphones and tablets without compromising security. These enhancements will significantly increase the system's effectiveness, scalability, and relevance in future digital learning ecosystems.

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